

Advanced Dynamics of Mechanical Systems

Assignment 2

Andrea Carlos Muratori (ID: 10829933)
Francesco Valerio Persio Pennesi (ID: 10821815)

June 12, 2026

1 Introduction

This report presents the results of the Experimental Modal Analysis (EMA) conducted on a 1:70 scale aeroelastic model of the *CityLife Canopy*. The objective of this work is to identify the system's modal parameters from experimental data and compare them against the predictions of a numerical Finite Element Model (FEM). The experimental data were acquired at the Wind Tunnel of Politecnico di Milano (GVPM). The structure was excited using a modal shaker with a “swept sine” force signal. The dynamic response, in terms of vertical velocity, was recorded across a grid of 202 measurement points distributed over the canopy surface using a scanning laser Doppler vibrometer (PSV Laser Doppler Vibrometer).

To perform the analysis, the following datasets were provided:

- The 202 experimental Frequency Response Functions (FRFs) relating the input force to the vertical displacement, computed using the H_1 estimator.
- The spatial coordinates of the laser measurement points.
- The reference FEM outputs, which include the numerical natural frequencies and mode shapes.

2 FRF Averaging and Preliminary Peak Detection

The first step in extracting the modal parameters consists of obtaining an initial estimate (first guess) of the system's resonance frequencies. For this purpose, the mean magnitude (absolute value) of all 202 experimental FRFs was computed within the frequency range of interest (2 – 8 Hz).

$$|\overline{G}(\Omega)| = \frac{1}{202} \sum_{j=1}^{202} |G_{jk}(\Omega)|. \quad (1)$$

The identification of local maxima is performed directly on this averaged spectrum. Specifically, a frequency is classified as a candidate peak if its magnitude strictly exceeds that of the 10 adjacent bins on either side, satisfying the condition $|\overline{G}|(i) > |\overline{G}|(i \pm l)$ for $l = 1, \dots, 10$. Figure 1 illustrates the magnitude spectrum of the 202 FRFs (in grey) superimposed on the mean curve (in blue), highlighting the identified peaks (red dots) that will be used as starting values for the identification process.

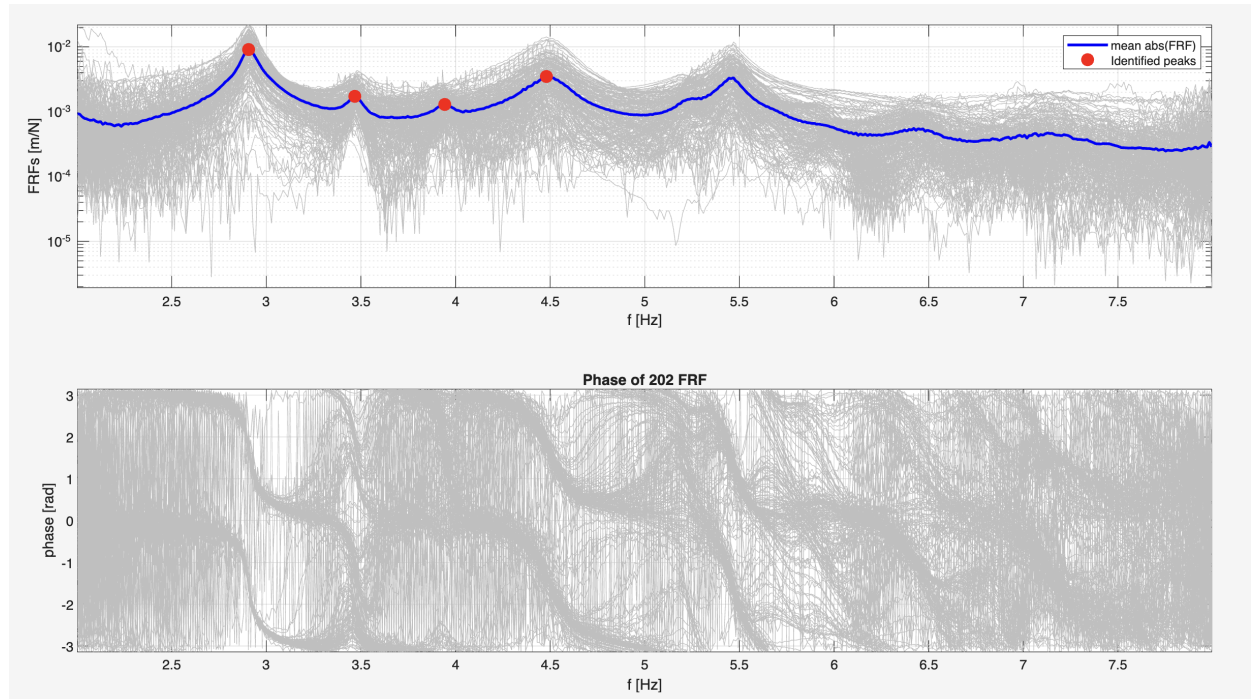


Figure 1

Table 1: Summary of the identified experimental natural frequencies. f_{exp} represents the resonance frequency extracted from the wind-tunnel FRFs.

Mode	1	2	3	4
f_{exp} [Hz]	2.9053	3.4668	3.9429	4.4800
$f_{fullexp}$ [Hz]	0.3472	0.4144	0.4713	0.5355

It can be observed that the resonance peaks are poorly defined and closely spaced; this will make any local approximation around the individual natural frequencies impossible because they are highly affected by each other.

3 Modal Parameter Identification (per mode)

For each targeted mode, the identification procedure involves selecting a subset of FRFs characterized by a high signal-to-noise ratio. Specifically, the FRFs with an amplitude in the top 15% at the initially guessed resonance frequency (representing points close to anti-nodal positions) are selected. These selected FRFs are then averaged in the complex domain. From this averaged anti-nodal FRF, the exact natural frequency, f_0 , is refined by locating the maximum of its absolute value:

$$f_0 = \arg \max_f |\bar{G}_{\text{anti-nodal}}(f)|$$

The corresponding modal damping ratio, h , is estimated by averaging the results obtained from two different methods applied to the same averaged FRF: the Half-Power Bandwidth method (h_{HP}) and the Phase Derivative method (h_{PD}).

$$h_{used} = \frac{h_{HP} + h_{PD}}{2}$$

Finally, the spatial mode shape is reconstructed by extracting the imaginary part of all 202 original FRFs evaluated at the refined resonance frequency f_0 , exploiting the proportional relationship at resonance:

$$\Phi(x_j) \propto \text{Im}[G_{jk}(f_0)]$$

The graphical results and the extracted parameters for the first four modes are presented in the sequence below.

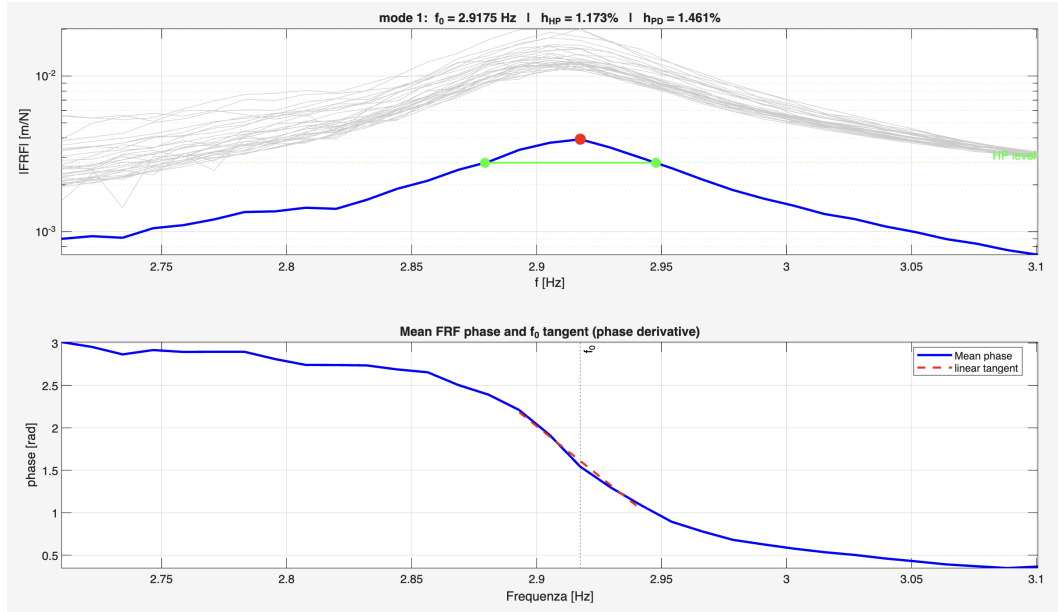


Figure 2: Mode 1 Identification: Figure 2 presents the identification results for the first mode. The refined natural frequency is located at 2.9175 Hz. As shown in the magnitude plot, the resonance peak is sharp and well-defined compared to the others. The unwrapped phase exhibits a clean, smooth transition, which leads to a reliable and consistent evaluation of the damping parameters.

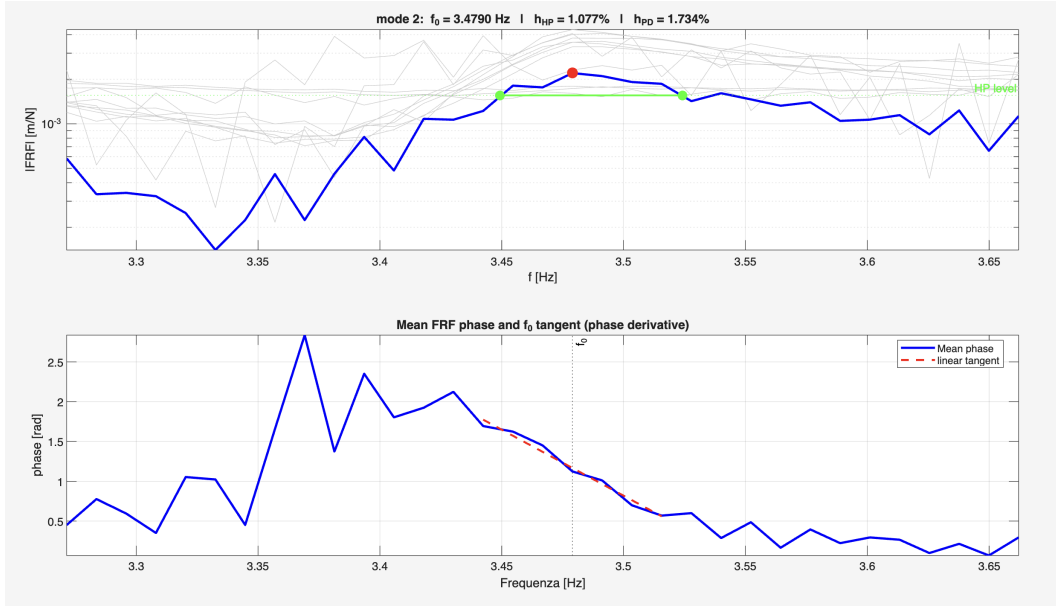


Figure 3: **Mode 2 Identification:** Figure 3 illustrates the parameter extraction for the second mode, identified at 3.4790 Hz. Unlike the first mode, the magnitude peak is flattened and less distinct, indicating strong modal overlap with adjacent frequencies. This interference causes visible distortion and noise in the phase plot, resulting in a significant discrepancy between the damping evaluated via the Half-Power method (1.077%) and the Phase Derivative method (1.734%).

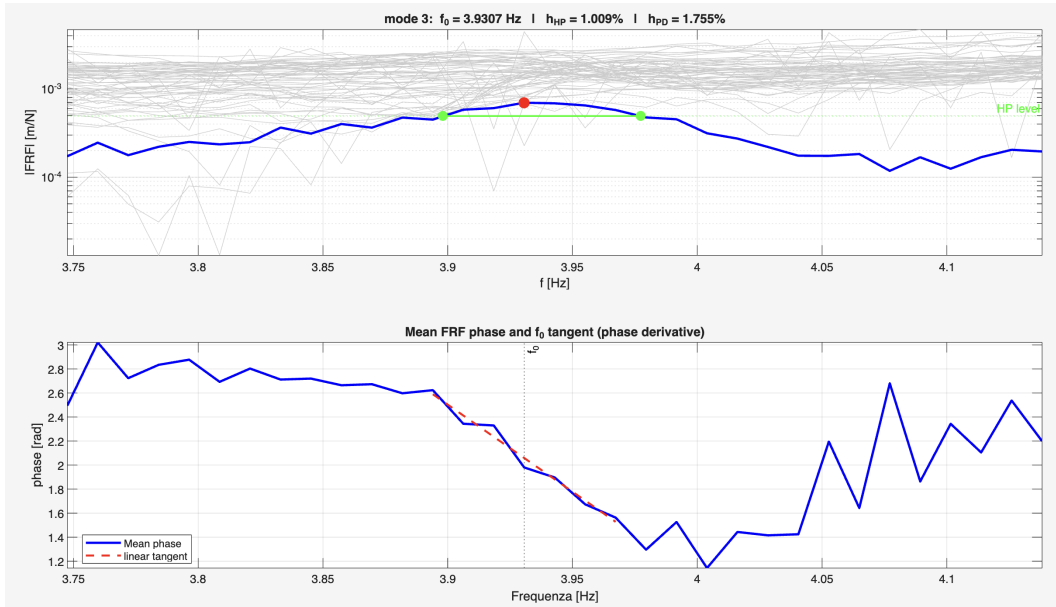


Figure 4: **Mode 3 Identification:** Figure 4 displays the results for the third mode, found at 3.9307 Hz. Similar to the second mode, this peak suffers from high modal density. The phase transition is notably jagged due to noise and modal crosstalk. Consequently, the damping estimates diverge considerably, with a Half-Power estimate of 1.009% and a much higher Phase Derivative estimate of 1.755%.

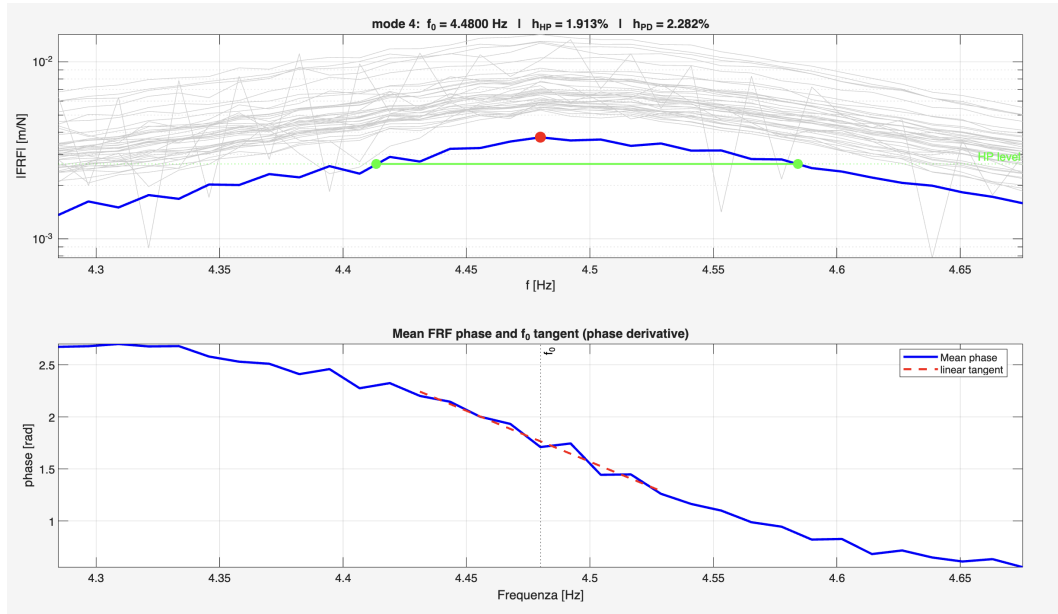


Figure 5: **Mode 4 Identification:** Figure 5 shows the identification of the fourth mode at 4.4800 Hz. The magnitude plot displays a wider resonance peak, which corresponds to the higher overall damping observed for this specific mode. While the phase plot is smoother than those of the central modes, some noise is still present along the curve.

4 Step 3 - Mode Shape Extraction and FEM Comparison

The mode-shape vector is therefore extracted as the imaginary part of all 202 FRFs evaluated at f_0 , then normalised so that the largest absolute value equals 1.

Since the laser measurement grid and the FEM nodes do not coincide, the experimental values are interpolated onto the FEM mesh with MATLAB's `scatteredInterpolant` (natural-linear interpolation).

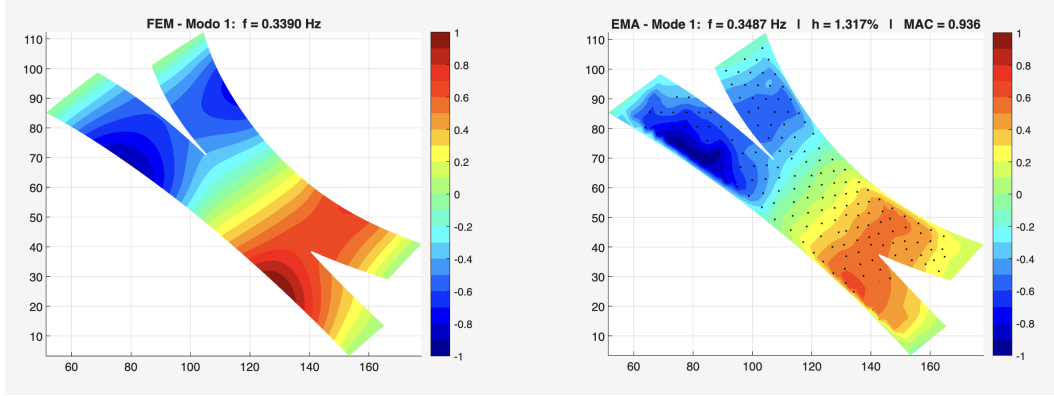


Figure 6: **Mode 1:** The first mode exhibits an excellent agreement between the numerical prediction ($f = 0.3390$ Hz) and the experimental data ($f = 0.3487$ Hz), achieving a highly satisfactory MAC value of 0.936. Visually, the EMA correctly captures the global, symmetric bending deformation of the canopy. This high spatial correlation perfectly aligns with the previous analysis: because this fundamental peak was sharp, well-isolated, and characterized by a clean phase transition, extracting the mode shape from the imaginary part of the FRF was highly effective and free from the effects of adjacent modes.

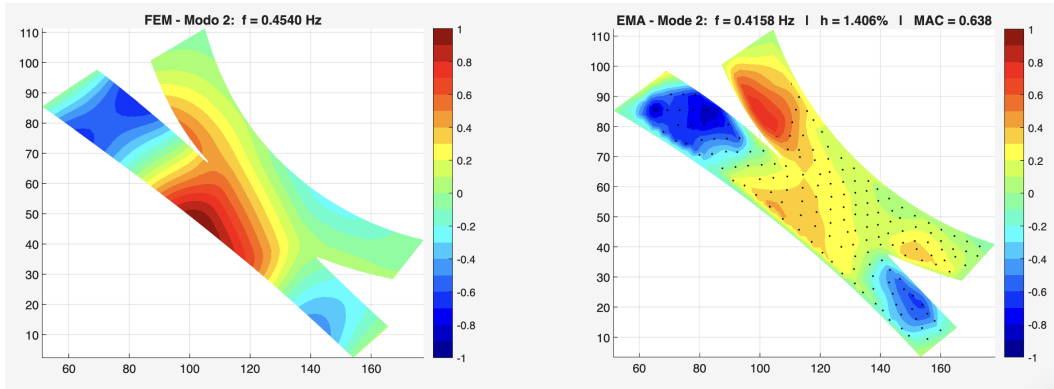


Figure 7: **Mode 2:** For the second mode, the spatial correlation drops significantly, yielding a MAC value of 0.638. Looking at the plots, while the numerical model (FEM) displays a very clear and orderly deformation, the shape extracted from the experimental data (EMA) appears differently distorted: the areas where the structure should positively deform (red zones) are highly compromised. This visual degradation is a direct consequence of the strong modal overlap discussed during the SDOF analysis. Because the resonance peak near 3.48 Hz is poorly defined and mixes heavily with the adjacent Mode 3, the extraction procedure using the imaginary part ends up capturing not only the shape of Mode 2 but also some of the energy from the neighboring mode.

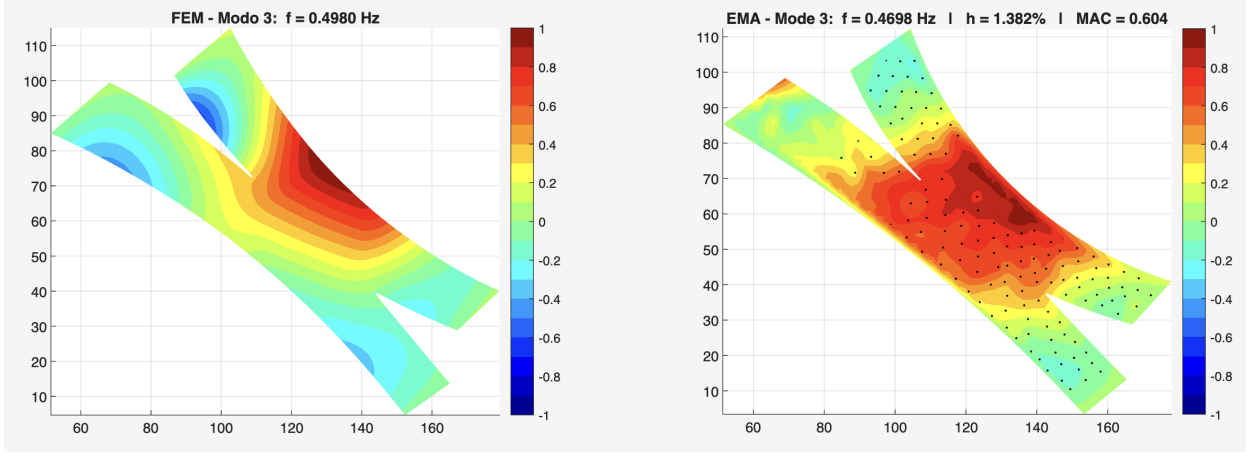


Figure 8: **Mode 3:** The third mode records the lowest spatial correlation among the investigated set, with a MAC of 0.604. The experimental mode shape struggles to reproduce the distinct negative-amplitude zones (blue zones) predicted by the FEM, meanwhile the positive amplitude zone appears significantly broadened. This result is entirely consistent with the previous analysis: the third peak exhibited the highest degree of phase distortion and the most ambiguous peak definition. The severe modal interference at this frequency makes the simple imaginary-part extraction method inadequate for isolating the pure mode shape.

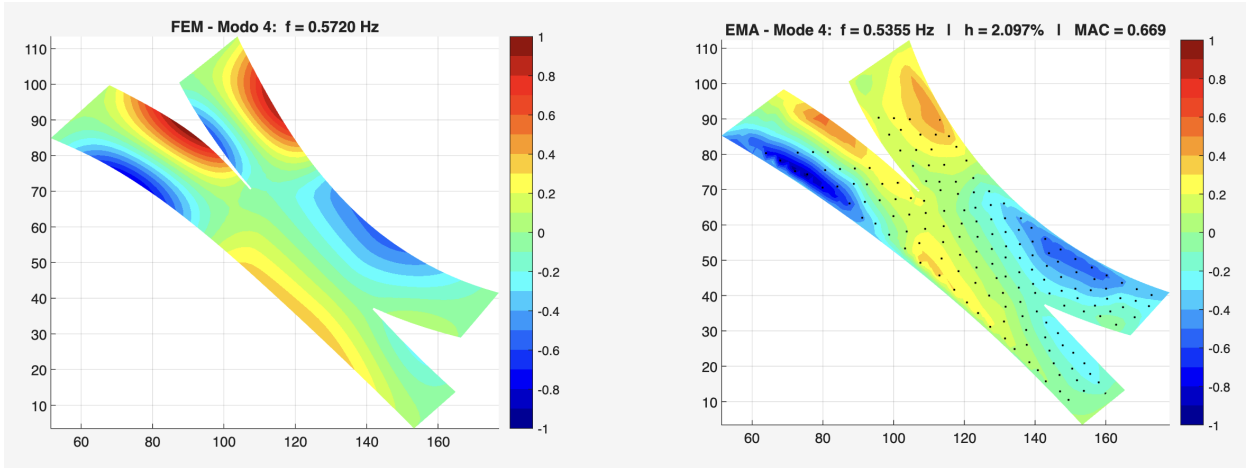


Figure 9: **Mode 4:** For the fourth mode, identified at $f = 0.5355$ Hz (EMA full-scale) compared to the numerical prediction of $f = 0.5720$ Hz (FEM), the spatial correlation shows a notable recovery, reaching a MAC of 0.669. While this value does not reach the optimal 0.80 threshold, the experimental mode shape successfully captures the main deformation pattern predicted by the numerical model. Crucially, unlike Modes 2 and 3, the identification of Mode 4 was significantly more robust due to its spectral isolation. This isolation decreased modal crosstalk, resulting in a cleaner phase transition and the most consistent damping estimates among all modes ($h_{HP} = 1.913\%$ and $h_{PD} = 2.282\%$). Therefore, the remaining spatial discrepancies and the moderate MAC are also caused by a lower signal-to-noise ratio due to the fact that this mode is stiffer than the others.

5 Cross-MAC Matrix

The Modal Assurance Criterion (MAC) quantifies the spatial correlation between two mode-shape vectors Φ_{exp} and Φ_{FEM} :

$$MAC(i, j) = \frac{|\Phi_{exp,i}^T \Phi_{FEM,j}|^2}{(\Phi_{exp,i}^T \Phi_{exp,i})(\Phi_{FEM,j}^T \Phi_{FEM,j})}. \quad (2)$$

A cross-MAC matrix is computed to detect potential mode-pairing errors between the numerical and experimental mode shapes.

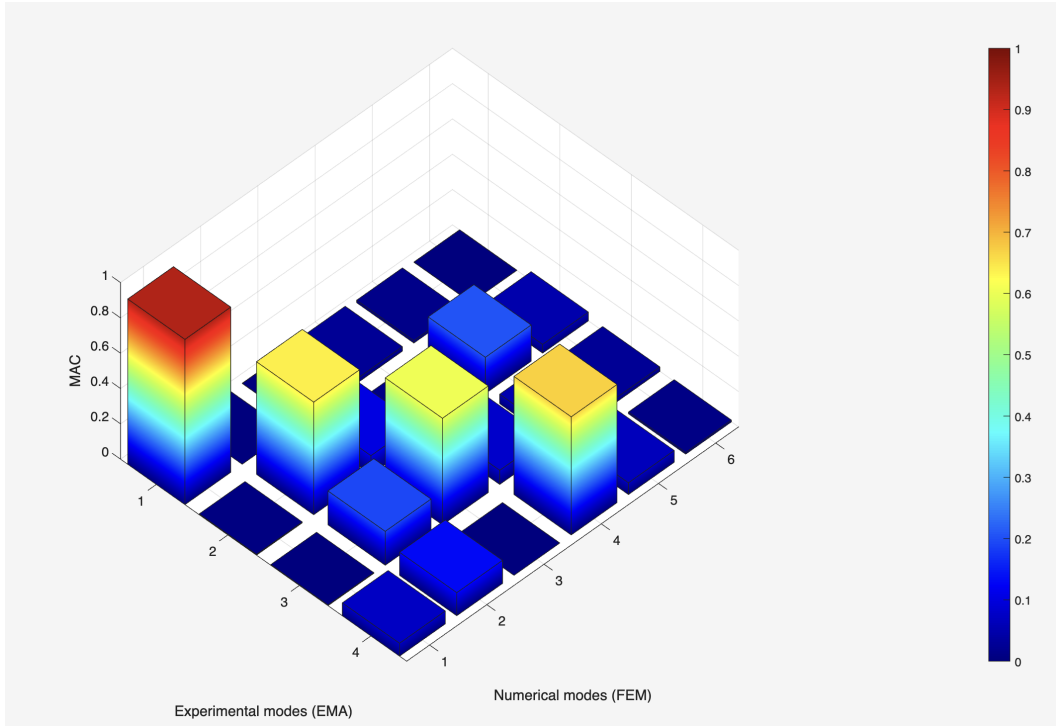


Figure 10: This global visualization effectively validates the earlier hypotheses regarding severe modal overlap. In an ideal, perfectly decoupled identification, all off-diagonal terms would be strictly zero. However, the matrix exhibits noticeable off-diagonal components. Most notably, the off-diagonal terms between the second and third modes definitively confirm their strong mutual coupling, which was heavily suspected during the SDOF extraction phase. Crucially, the expanded correlation analysis including numerical modes 5 and 6 reveals additional insights into the poor spatial reconstruction of EMA Mode 2. The 3D plot highlights a distinct off-diagonal correlation between Experimental Mode 2 and Numerical Mode 5.

6 Results

Table ?? summarizes the complete set of identified modal parameters for the first four structural modes. The natural frequencies are reported at both the experimental model scale (as directly extracted from the wind-tunnel FRFs) and the corresponding full scale. The conversion to full-scale frequencies was performed by applying the Froude scaling factor of $\sqrt{1/70} \approx 0.1195$.

Mode	f_{exp} [Hz]	f_{full} [Hz]	f_{FEM} [Hz]	Err. [%]	h_{HP} [%]	h_{PD} [%]	h_{used} [%]	MAC
1	2.9175	0.3487	0.3390	+2.86	1.173	1.461	1.317	0.936
2	3.4790	0.4158	0.4540	-8.41	1.077	1.734	1.406	0.638
3	3.9307	0.4698	0.4980	-5.66	1.009	1.755	1.382	0.604
4	4.4800	0.5355	0.5720	-6.39	1.913	2.282	2.097	0.669

7 Conclusion

The experimental modal analysis conducted on the 202-point dataset has successfully provided a robust characterization of the structure’s dynamic properties within the 2 – 8 Hz range. The investigation confirmed that while the fundamental mode is well-isolated and exhibits high spatial correlation with the numerical model, the intermediate modes (2 and 3) are characterized by significant modal density and coupling. The integration of damping estimates and the evaluation of the Modal Assurance Criterion (MAC) matrix provided a comprehensive overview of the system’s behavior, highlighting both the strengths of the experimental approach and the limitations imposed by modal crosstalk.